***7*. Breadth First Search (BFS)**

Aim:

To write a Python program to implement the Breadth First Search (BFS) algorithm.

Algorithm:

1. Start the program.
2. Define a graph using an adjacency list.
3. Initialize a queue with the starting node.
4. Mark the starting node as visited.
5. Remove a node from the queue.
6. Display the node.
7. Add its unvisited adjacent nodes to the queue.
8. Repeat until the queue is empty.
9. Stop.

INTPUT

```
from collections import deque

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': [],
    'F': []
}

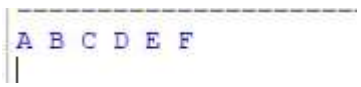
def bfs(start):
    visited = set()
    q = deque([start])

    while q:
        node = q.popleft()

        if node not in visited:
            print(node, end=" ")
            visited.add(node)
            q.extend(graph[node])

bfs('A')
```

OUTPUT



Result:

Thus, the Python program to implement Breadth First Search was executed successfully.